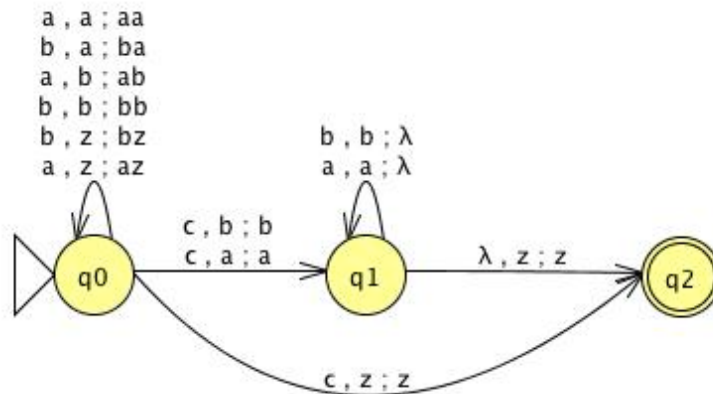


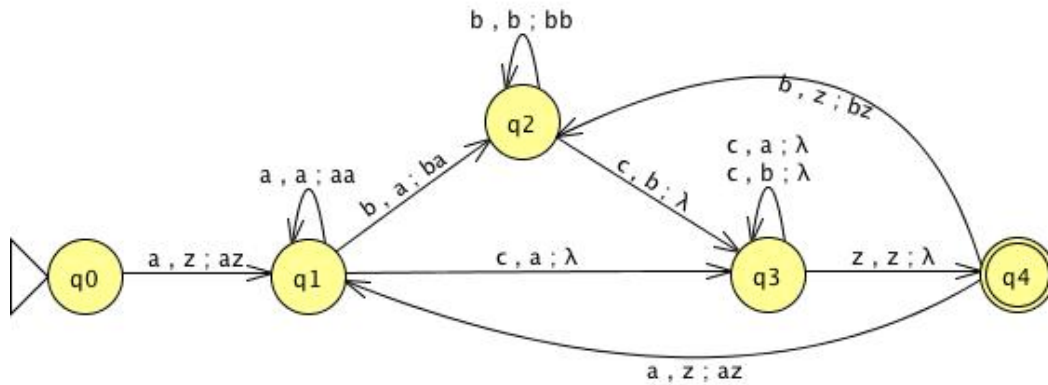
4) Construct nPDA's that accept the following languages on $\Sigma = \{a, b, c\}$.

b)

$L = \{wcw : w \in \{a, b\}^*\}$



c) $L = \{a^n b^m c^{n+m} : n \geq 0, m \geq 0\}$

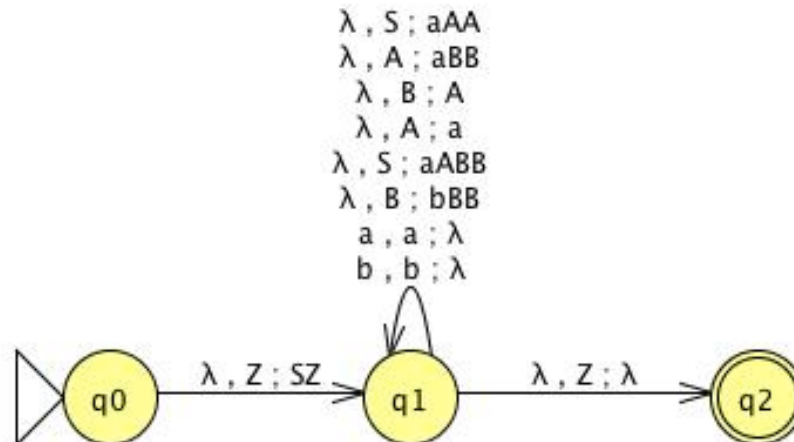


5) Construct an nPDA corresponding to the grammar

$S \rightarrow aABB|aAA,$

$A \rightarrow aBB|a,$

$B \rightarrow bBB|A$



7) Show that the following languages on $\Sigma = \{a, b, c\}$ are not context-free

c) $L = \{a^n b^j c^k : k = jn\}$

Assume, for a contradiction, that L is regular so that there is a pumping length $m > 0$.

Choose $w = a^m b^m c^{2m}$ so that $w \in L$ and $|w| \geq m$.

Let $w = uvxyz$ be the decomposition of w given by the pumping lemma.

- ✓ $|vxy| \leq m$.
- ✓ $|vy| \geq 1$.
- ✓ $uv^i xy^i z \in L$ for all $i \geq 0$.

Choose $i = 0$ so that $w_i = uv^0 xy^0 z$.

First case, $w_i = a^{m-|vy|} b^m c^{m^2}$ and $w_i \notin L$. Second case, $w_i = a^{m-|v|} b^{m-|y|} c^{m^2}$ and $w_i \notin L$. Third case, $w_i = a^m b^{m-|vy|} c^{m^2}$ and $w_i \notin L$. Forth case, $w_i = a^m b^{m-|v|} c^{m^2-|y|}$ and $w_i \notin L$. Fifth case, $w_i = a^m b^m c^{m^2-|vy|}$ and $w_i \notin L$.

Choose any $i > 0$, cases where v or y contain a sequence of ab 's or bc 's where w_i would be $a^+ ab^+ ba^+ b^+$ or $a^+ ab^+ bc^+ cb^+ c^+$.

All cases contradict our assumption and hence, L is not context free grammar.

8) Determine whether or not the following languages are context-free.

a) $L = \{a^n w w^R a^n : n \geq 0, w \in \{a, b\}^*\}$

Assume, for a contradiction, that L is regular so that there is a pumping length $m > 0$.

Choose $W = a^m a^m b^m b^m a^m a^m$ where $w = a^m b$ and $w^R = b a^m$ and so that $W \in L$ and $|W| \geq m$.

Let $W = a^m a^m b^m b^m a^m a^m \rightarrow W = a^{2m} b^{2m} a^{2m} = a^p b^p a^p$ where $p = 2m$.

Let $W = uvxyz$ be the decomposition of W given by the pumping lemma.

- ✓ $|vxy| \leq m$.
- ✓ $|vy| \geq 1$.
- ✓ $uv^i xy^i z \in L$ for all $i \geq 0$.

Based on the above characteristics we can have the following cases:

First, v or y contain at least an "a" in the first block of a 's. If we pump up or down, it will result in a mismatch with the second block of a 's because we cannot have any a 's from the second block in v or y .

Second, v or y contain at least an "a" in the last block of a 's. As in the first case v or y cannot reach all a 's in the beginning of the string.

Third, v and y contain b 's; If we pump up, it would violate the $n \geq m$ constraint since the number of b 's will exceed the number of a 's in each part because $|vy| > 0$.

Finally, all cases contradict our assumption and hence L is not context free grammar.